

Assignment #1

Write on your General Science Lecture Notebook (Small)

A. Identify the Newton's Law of Motion described by each given equation.

- _____ 1. When a car stops suddenly, passengers lurch forward because their bodies tend to continue moving at the same speed until restrained by a seatbelt
- _____ 2. When swimming, pushing water backward propels the swimmer forward
- _____ 3. A spinning fan continues to rotate briefly after being turned off due to inertia
- _____ 4. A soccer ball stays still until kicked, and a rolling ball eventually stops because of friction
- _____ 5. Pushing a shopping cart is easier than pushing a car because the car has more mass, requiring more force to achieve the same acceleration
- _____ 6. Rockets launch upward because gases are expelled downward, producing thrust
- _____ 7. A book lying on a table remains stationary until someone pushes it
- _____ 8. Pedaling harder on a bicycle increases its speed because more force is applied
- _____ 9. Rocket engineers calculate the thrust needed to lift a spacecraft off Earth using this law
- _____ 10. Walking involves pushing the ground backward with your feet, which moves your body forward

B. Solve the problem

Read each problem carefully and solve for the required quantity. Use the appropriate formula, show all necessary computations, and write your final answer with the correct unit.

1. A soccer ball with a mass of 0.5 kg is kicked with a force of 15 N. Determine the acceleration of the ball.

2. A force of 60 N acts on a cart, causing it to accelerate at 3 m/s^2 . What is the mass of the cart?

3. A 12-kg object accelerates at 2.5 m/s^2 . How much force is applied to the object?

4. A motorcycle with a mass of 150 kg accelerates at 4 m/s^2 . Calculate the net force acting on the motorcycle.

5. A 25-kg sled is pulled with a force of 100 N across a smooth surface. What is its acceleration?

6. A truck experiences a net force of 4,000 N and accelerates at 2 m/s^2 . Find the mass of the truck.

7. A 70-kg runner pushes off the ground and accelerates at 1.5 m/s^2 . What force causes the runner's acceleration?

8. A 2-kg toy car accelerates at 6 m/s^2 . Determine the net force acting on the toy car.

Challenge Problems

9. Two students push a 20-kg cart in the same direction. One exerts a force of 40 N and the other exerts a force of 30 N. What is the cart's acceleration?

10. A 1,200-kg car accelerates from rest at 3 m/s^2 . What net force is required to produce this acceleration?

These problems progress from basic substitution to slightly more complex applications of